

# Two theories about the other half

Participation explains roughly half of the elite chess rating gap. The debate almost never asks what the other half is made of. Two answers get asserted constantly, both are testable on public data, and both fail.

FIDE standard rating lists, April 2016 to April 2026. 716,730 active players across six annual cohorts.

The participation argument has been running for fifteen years and has hardened into two camps: one says sample size explains the gap, the other says it doesn't. Both camps quote a percentage and stop. Nobody goes on to test what the unexplained remainder actually contains, even though the two most popular candidates leave fingerprints in the same public dataset the percentage came from.

So here they are, tested.

## 1 Theory one: women drop out earlier

The claim is that women leave competitive chess sooner, so the female pool is not merely smaller but younger and less experienced at every rating. If true, it would explain a residual gap without appealing to ability.

**Test.** Take every player active in April of each year from 2016 to 2021. Ask whether they are still on the list and still active five years later. Compare women with men matched on rating band and cohort, because the raw comparison is meaningless: the median woman in these cohorts is 20 and rated 1700, the median man is 42 and rated 1809, and both age and rating independently predict leaving.

Raw, women survive at 48.5% against 55.3% for men. Matched, almost all of that composition effect disappears and what remains sits in exactly one place:

Odds of still being active five years later, women vs men, matched on rating and cohort.

| AGE AT BASELINE | MEN   | WOMEN | ODDS RATIO  | 95% CI       |
|-----------------|-------|-------|-------------|--------------|
| under 13        | 63.3% | 68.1% | <b>1.37</b> | [1.25, 1.50] |
| 13-17           | 43.5% | 44.2% | <b>1.19</b> | [1.15, 1.24] |
| 18-24           | 48.9% | 41.5% | <b>0.86</b> | [0.83, 0.89] |
| 25-39           | 54.0% | 52.0% | 0.96        | [0.92, 1.00] |
| 40-59           | 61.5% | 60.5% | 1.01        | [0.96, 1.07] |
| 60+             | 55.6% | 53.8% | 0.99        | [0.91, 1.07] |

Girls are *more* likely than matched boys to still be playing. The reversal happens between 17 and 18 and is gone again by 25. It holds in all six cohorts separately and widens with the horizon, so it is a steady leak in early adulthood rather than a pandemic artefact or a cliff.

**But it is a club-level effect, and that is what kills it as an explanation.** Hold age at 18-24 and raise the strength requirement:

Ages 18-24 only, by baseline rating.

| BASILINE RATING | ODDS RATIO  | 95% CI       | N WOMEN |
|-----------------|-------------|--------------|---------|
| 1400+           | 0.86        | [0.83, 0.89] | 14,258  |
| 1800+           | 0.91        | [0.86, 0.96] | 5,810   |
| 2000+           | 1.05        | [0.96, 1.15] | 2,388   |
| <b>2200+</b>    | <b>1.28</b> | [1.05, 1.55] | 674     |

Above 2000 the gap closes. Above 2200 it reverses. The elite comparison that produces the residual is drawn from players rated far above 2200, and in precisely that population women are retained at least as well as men.

#### VERDICT

Differential drop-out is real, well identified, and located at club level between 18 and 24. It is a finding about how the participation ratio gets made, not about the gap that ratio fails to explain. It cannot be the missing half.

## 2 Theory two: women get fewer chances to play

The second claim is about opportunity: fewer invitations, fewer events, less competitive practice, so less improvement per year of nominal membership.

**Test.** Count rated games actually played over twelve months, comparing men and women in the same rating band.

Rated games played over 12 months, active players, matched on rating band.

| RATING BAND | MEDIAN, MEN | MEDIAN, WOMEN | MEAN, MEN | MEAN, WOMEN |
|-------------|-------------|---------------|-----------|-------------|
| 1800-2000   | 8           | <b>14</b>     | 14.9      | 25.3        |
| 2000-2200   | 11          | <b>22</b>     | 20.6      | 32.5        |
| 2200-2400   | 17          | <b>44</b>     | 29.9      | 47.7        |
| 2400+       | 37          | <b>55</b>     | 47.6      | 57.9        |

The effect runs the wrong way, in every band, and widens with strength. At 2200-2400 the median woman plays roughly two and a half times as many rated games as the median man of the same strength.

#### VERDICT

Whatever the opportunity story is, it is not a story about volume of competitive play. Strong women play more chess than equally strong men, not less.

## 3 What this does and does not show

Two named candidates, tested on the same data that produced the percentage, both eliminated at the strength level where the residual actually lives. That is a real contribution, and it is worth more than another estimate of the percentage itself.

What it is not is an argument that the residual reflects ability. Eliminating two explanations does not promote a third. The remaining candidates include developmental effects that land long before anyone appears on a rating list, differences in who enters the rated population at all, coaching and training quality as distinct from playing volume, and measurement artefacts in the rating lists themselves. None of those is tested here.

Three limits worth stating rather than burying. The 2200+ retention cell rests on 674 women, so the reversal should be read as "no deficit," not as an advantage. Matching on rating and age is not matching on selection: a rated woman is a more selected individual than a rated man at every level, because the filter into the rated population is tighter, and nothing here controls for that.

And the third cuts across both results at once. Women's-only events and titles give strong women both a reason to keep competing and a supply of tournaments with no male equivalent. That is a single plausible mechanism which could produce the retention reversal *and* the games-played reversal simultaneously. It does not rescue either theory, since both were claims about deficit and the data shows surplus, but it does mean these two results are less independent than they look.

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## METHOD

Retention: six annual cohorts of players active in April 2016 through April 2021, each tracked five years. Effects are Mantel-Haenszel odds ratios stratified by rating band and cohort; a logistic model with the same controls agrees to two decimal places. A baseline rating floor of 1400 is applied throughout, because FIDE raised the bottom of the published list from 1000 to 1400 during 2024-25 and women are more than twice as likely to sit below it, so without the floor administrative removal would be counted as attrition and counted unevenly.

Games played: active players, 12 monthly snapshots May 2025 to April 2026, games summed per player, compared within rating bands.

Both analyses use FIDE standard rating lists only. Players holding rapid or blitz ratings without a standard rating do not appear.